SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Rohit Kumar S	**Reg. No.:**	192312214
Section / Batch:	2023	**Date:**	

Code of Conduct

I **Rohit Kumar S** (192312214) (*Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____**rohit**_____

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.



ITA 0328- MOBILE COMPUTING

Challenges in Mobile Computing and Cellular Networks

Guided By
Dr. Veeramani

Presented By
Rohit Kumar S (192312214)

Introduction

Mobile computing enables users to access data, applications and services while moving across locations and networks.

Uses portable devices, wireless links and distributed computing resources.

Cellular networks connect devices through radio access networks and a core network.

Technologies such as 4G LTE and 5G support voice, messaging, broadband and connected devices.

Performance depends on radio conditions, network capacity, mobility and security.

Mobility

Maintain service while users move.

Connectivity

Provide reliable access over wireless links.

Continuity

Keep applications responsive and available.

Core Technical Challenges

Limited bandwidth

Radio spectrum is finite. Many users and high-demand applications compete for capacity.

Variable signal quality

Distance, obstacles, interference and fading can reduce throughput and reliability.

Latency and congestion

Busy cells and network processing can delay interactive services and real-time traffic.

Device constraints

Battery life, processing capability, heat and storage limit mobile-device performance.

Mobility, Handover and Coverage

Why mobility is difficult

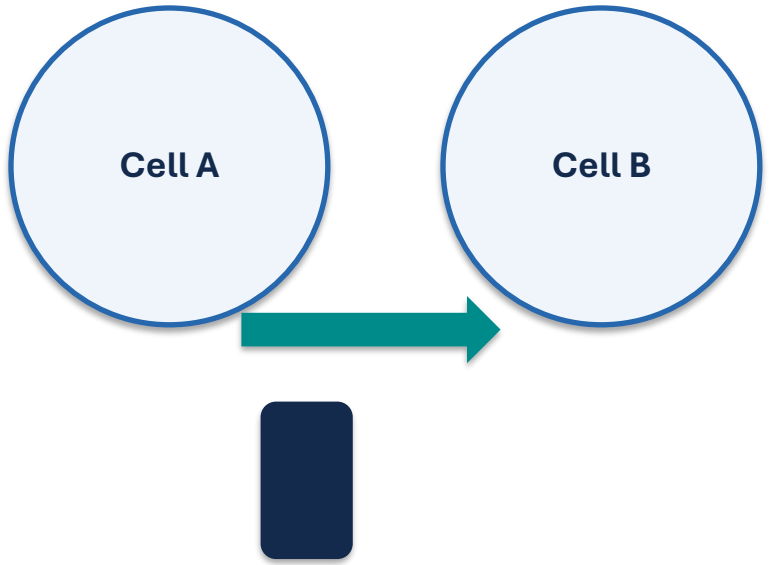
Handover must transfer an active connection between cells with minimal interruption.

Fast movement, overlapping coverage and fluctuating signal strength complicate decisions.

Coverage gaps can occur indoors, in rural areas, tunnels and crowded venues.

Roaming across operators or technologies can introduce authentication and compatibility issues.

Illustrative cell handover



Device moves from Cell A to Cell B

Security and Privacy Challenges

Wireless access and always-connected devices increase the need for strong, continuous protection.

Identity & access

Weak credentials or flawed authentication can expose accounts and services.

Wireless threats

Eavesdropping, spoofing and rogue access points may compromise communications.

Privacy exposure

Location, usage patterns and personal data can be collected or misused.

Device & app risk

Unpatched software, unsafe apps and lost devices can create entry points.

Priority measures: encryption • multi-factor authentication • timely updates • least-privilege access

Network Management and Emerging Demands

Capacity planning: dense urban traffic and video, gaming and cloud services increase demand.

Quality of service: applications have different needs for throughput, delay and reliability.

Energy efficiency: networks must balance performance with power use at devices and base stations.

Interoperability: devices, standards, legacy systems and operators must work together.

New use cases: IoT, connected transport and industrial systems raise scale and reliability requirements.

Scale

Support many devices per area.

Efficiency

Use spectrum and energy wisely.

Reliability

Meet application-specific service needs.

Conclusion and Way Forward

Mobile computing and cellular networks are essential to modern digital services, but their performance is shaped by interconnected challenges.

Improve coverage and capacity through careful radio planning and efficient spectrum use.

Use intelligent mobility management and quality-of-service mechanisms.

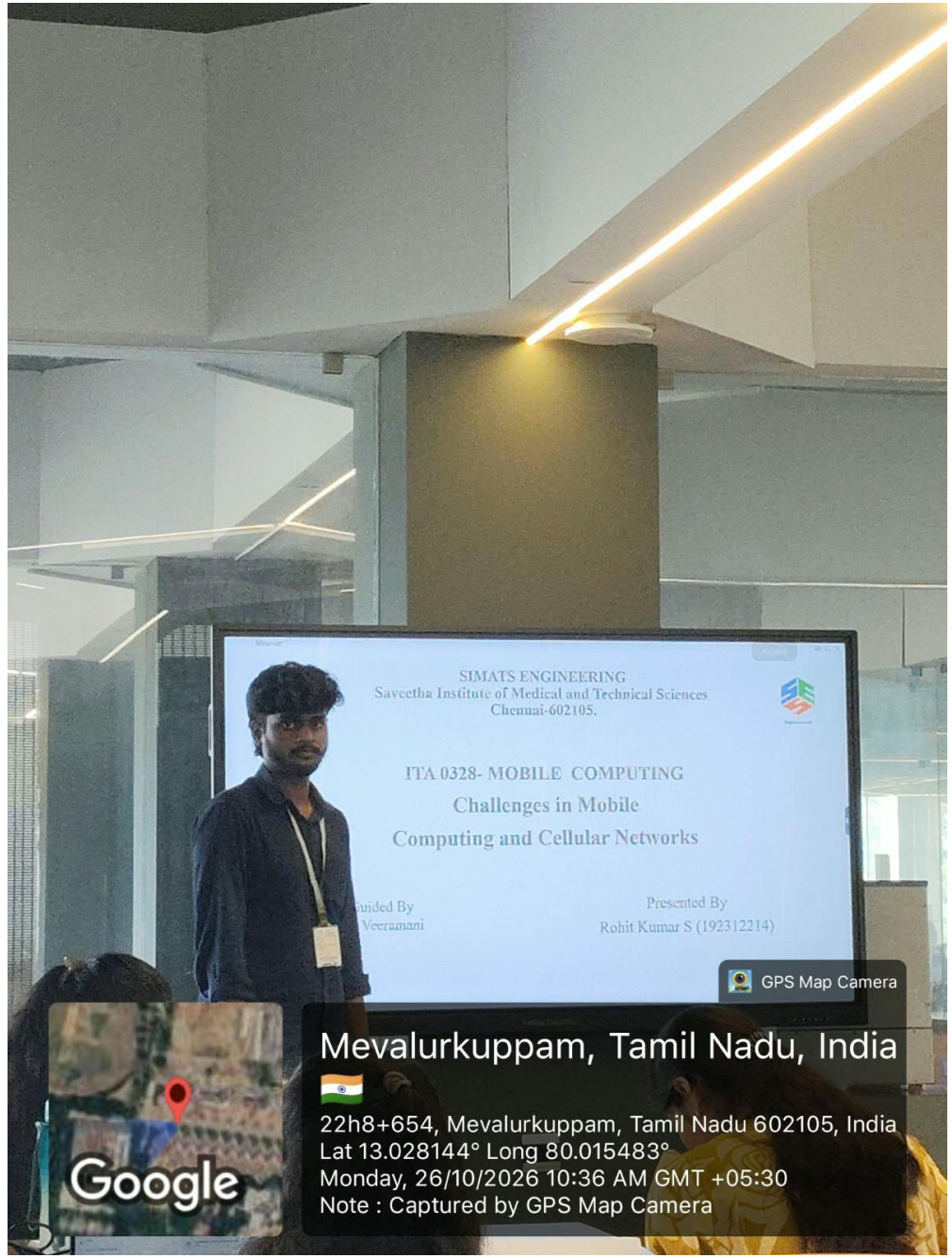
Strengthen privacy and security through encryption, authentication and regular updates.

Design energy-efficient devices and resilient network infrastructure.

Key takeaway

Reliable mobile services require a balance of speed, coverage, security, efficiency and user experience.

Thank you


SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.
ITA 0328- MOBILE COMPUTING
Challenges in Mobile
Computing and Cellular Networks
Presented By
Rohit Kumar S (192312214)
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22h8+654, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.028144° Long 80.015483°
Monday, 26/10/2026 10:36 AM GMT +05:30
Note : Captured by GPS Map Camera
Google